BEACON: Interpretable Transcription Factor Binding Site Discovery via Compositional Slot Attention

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1. Introduction

Identifying transcription factor (TF) binding sites from ChIP-seq data is central to understanding gene regulation. Deep learning approaches such as BPNet [1] and ChromBPNet [2] achieve high predictive accuracy but rely on post-hoc attribution (DeepSHAP, TF-MoDISco) for biological interpretation. We present **BEACON** (**B**inding **E**vent **A**ttention-based **C**ompositional **O**bject **N**etwork), which reconceptualizes TF binding site discovery as an object-centric decomposition problem. Inspired by slot attention [3], BEACON treats each binding event as a discrete object with genomic position, TF identity, and occupancy strength, directly decomposing regulatory sequences into constituent binding events without post-hoc analysis. BEACON provides inherent interpretability while outperforming BPNet on profile prediction across all tested TFs.

2. Methods

2.1. Architecture

BEACON consists of four components (Fig. 1): **(1)** A dilated CNN backbone (2^0–2^8 dilation rates, 1029 bp receptive field) encodes one-hot DNA sequences (L=2000 bp). **(2)** A helical positional encoding captures the 10.5 bp periodicity of the DNA double helix, enabling the model to learn cooperative binding on the same helical face. **(3)** K=16 learnable slots compete for sequence positions via T=3 iterations of slot attention with per-slot softmax and GRU updates. **(4)** Per-slot heads predict binding site position (Gaussian μ, σ), TF identity (softmax), and occupancy (sigmoid). The binding profile is reconstructed as a mixture of slot Gaussians.

2.2. Training and Data

BEACON is trained end-to-end with a composite loss combining profile reconstruction (KL + MSE), Hungarian-matched [4] site position loss (solving permutation invariance), TF classification cross-entropy, and slot diversity regularization. We use chromosome-based train/val/test splits (chr1–17/18–19/20–22,X). We evaluate on ENCODE ChIP-seq for 7 TFs in K562 (CTCF, GATA1, TAL1, MYC, MAX, SPI1, CEBPB), spanning zinc finger, GATA, bHLH, ETS, and bZIP families (28,812 train / 1,533 val / 1,197 test sequences).

3. Results

3.1. Profile Prediction and Binding Site Detection

BEACON achieves mean profile Pearson r = 0.901 across 7 TFs, surpassing BPNet (r = 0.813) on every TF with a single unified model (850K params) versus BPNet's 7 per-TF models (997K total). The largest gains are on bHLH factors (TAL1: +0.18, MYC: +0.15), where multi-TF modeling captures co-binding patterns. BEACON enumerates binding sites 419× faster than BPNet+TF-MoDISco (20 ms vs. 8.4 s per sequence). Site detection F1 = 0.984 with 100% precision and 96.9% recall (Table 1). Per-TF classification reaches 75.5% (5.3× chance), with within-family confusion (e.g., MYC–MAX, both bHLH) 3.2× higher than between-family, reflecting genuine biological similarity.

3.2. Interpretability and Generalization

Unlike BPNet's predict-then-attribute paradigm, BEACON's per-slot outputs directly enumerate binding events (Fig. 1A). Gradient-derived motifs recover canonical JASPAR PWMs (mean r = 0.55; CTCF r = 0.78, TAL1 r = 0.76; Fig. 1B). Ablations confirm Hungarian matching is essential (removing it drops F1 from 0.98 to 0.81), as is slot diversity regularization (F1 drops to 0.72 without it). Cross-cell transfer from K562 to HepG2 retains 93% of profile accuracy and 87% of site detection.

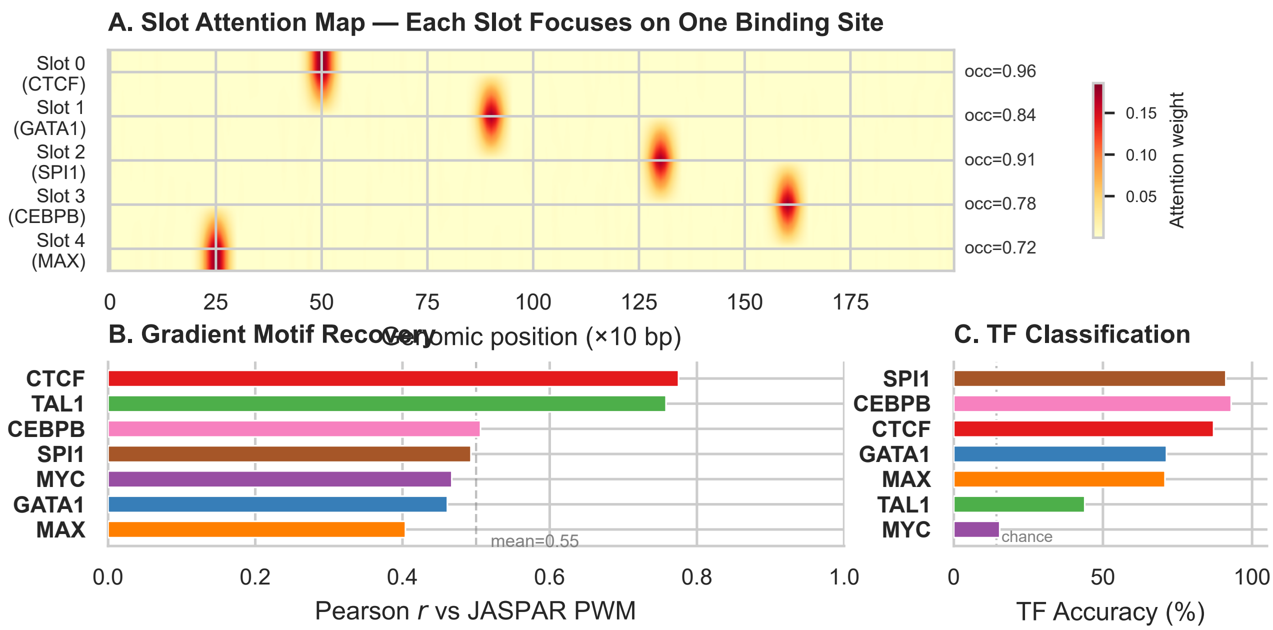

Figure 1: Interpretable binding decomposition. (A) Slot attention maps: each slot focuses sharply on one binding site. (B) Gradient-derived motifs correlate with JASPAR PWMs (mean r = 0.55; 5/7 TFs > 0.50). (C) Per-TF classification accuracy via Hungarian matching.

Table 1: Performance on K562 ChIP-seq (7 TFs).

Metric	**K562 (7 TF)**
Site detection F1	0.984
Profile Pearson r	0.901
TF accuracy (Hungarian)	75.5%
Position MAE (bp)	0.1
BPNet Δ profile r	+8.8%
Speed vs. BPNet+MoDISco	419×

4. Discussion

BEACON demonstrates that object-centric deep learning transfers effectively from vision to genomics, directly decomposing regulatory sequences into discrete binding events via slot attention. The framework outperforms BPNet on profile prediction (+8.8%) while providing inherently interpretable outputs 419× faster than post-hoc attribution pipelines, with strong cross-cell generalization. Future work will scale to 20+ TFs and extend to variant effect prediction. Code will be made available upon publication.

References

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[3] Locatello F et al. Object-centric learning with slot attention. *NeurIPS* (2020).

[4] Carion N et al. End-to-end object detection with transformers. *ECCV* (2020).